

Supplementary Material

An Event-Dated Corpus of US Trademark Prosecution and a Two-Sided Measure of Vocabulary Position

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This supplement carries the material the article summarizes: registration-step detail (§S1), the term-versus-theme representation comparison (§S2), the burn-in profile (§S3), and the full robustness treatment (§S4). Numbered references to sections, tables and figures without an S prefix point into the article.

S1 Registration detail

S1.1 Registration completion peaks at middling atypicality

Across the scored debut filings (57.6% base rate), completion is an inverse-U in *atypicality*: it climbs from 46.6% at the least atypical end to about 58% in the middle and falls back to 53.4% at the most atypical (Figure S2, panel C). The same shape appears on the past-facing level alone, more mildly, running 54.0% at the bottom quintile to 58.4–59.1% in the middle and 57.7% at the top (Figure S1A).

The signed axis behaves differently. Completion on L falls to a trough of 51.9% just above the median and then rises steadily to 58.0% at the leading end. That trough is composition rather than timing. Lead magnitude correlates with atypicality at +0.31, so the filings near zero lead are disproportionately the low-atypicality filings, and those complete poorly for the reason above. Splitting by atypicality confirms it: the trough is absent in the upper three fifths and survives only in the lowest. In a linear specification the signed axis enters positively — leading filings register slightly more often — but monotonically rather than through an interior peak: what curvature exists at registration belongs to how far a filing’s language sits from its class, not to which side of it the filing sat.

The obvious objection is that unfamiliar language is simply refused more often, so the measure recovers nothing beyond familiarity. Completion falls at *both* ends, so the most class-typical filings also complete less than the middle, which familiarity cannot explain; and the curvature is specific to the unsigned axis, the signed axis showing no interior peak at all.

Refiling after abandonment is itself mildly related to vocabulary position. Refiling rates are flat in atypicality (a middle-minus-tails difference of -0.06pp on a 3.6% base) and peak in the middle of the lead distribution (4.15%), falling to 2.6% among the most lagging abandoned applications and 3.4% among the most leading.

S1.2 Self-filed and professionally filed applications

Counsel of record is observed on every filing. Represented applications are 73% of filings over 1995–2018, falling from 85% in 1995 to 68% in 2018 as online self-filing spread. The two populations file

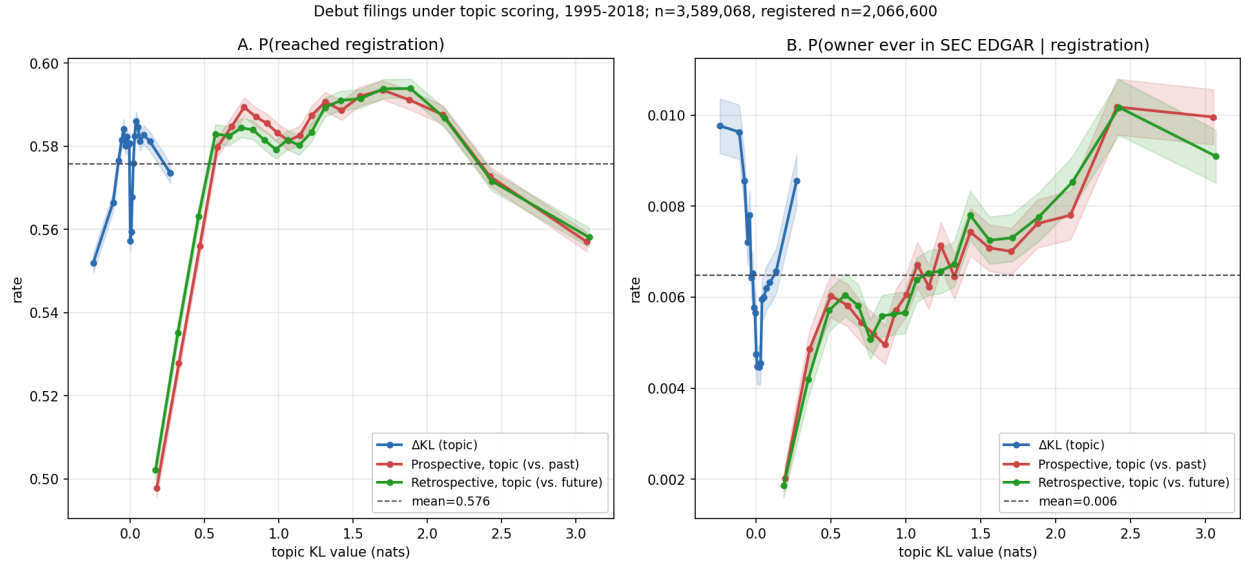


Figure S1: Outcomes for debut filings (owner’s first-ever filing) across all 45 Nice classes under theme scoring, filing years 1995–2018, in quantile bins with 95% intervals, quantiles cut on the pooled distribution. $n = 3,589,068$; registered subset $n = 2,066,600$. *A*: P(reached registration). *B*: P(owner ever in SEC EDGAR, the Commission’s public filing system | registration).

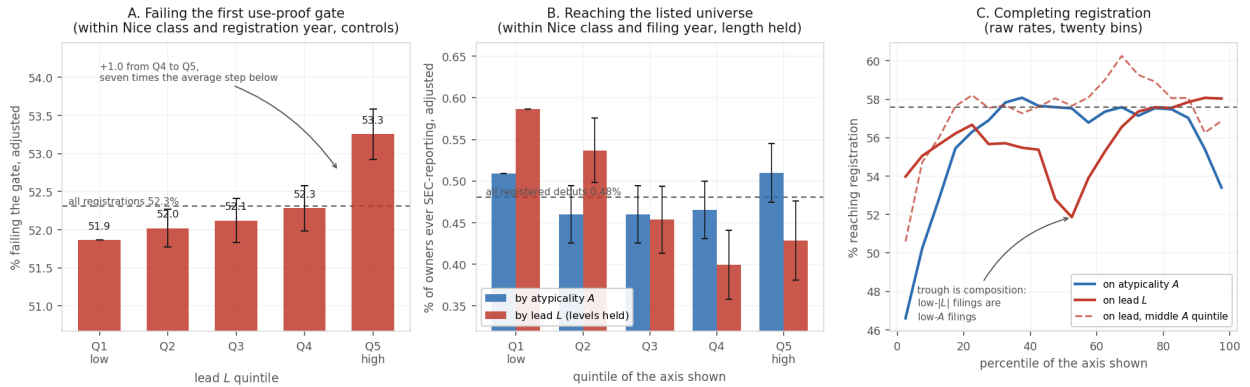


Figure S2: Quintile profiles for the three outcomes. *A*: the share of registrations failing the five-year proof of continued use, by lead quintile, adjusted within Nice class and registration year with the drafting and filer controls of Table 3; the dashed line is the rate for all registrations. The profile rises gently and monotonically through the fourth quintile and then steps up: about seven tenths of the top-to-bottom gap is the single step from the fourth quintile to the fifth, seven times the average step below it. *B*: the share of first-time filers whose owner ever reaches SEC reporting, by quintile of atypicality and of lead, adjusted within Nice class and filing year with description length held. *C*: raw registration rates for debut filings in twenty bins, cut on the pooled distribution — an inverse-U on atypicality, and on lead a trough just above the median that is compositional, absent once atypicality is held in its middle fifth (dashed). Horizontal line at the pooled rate.

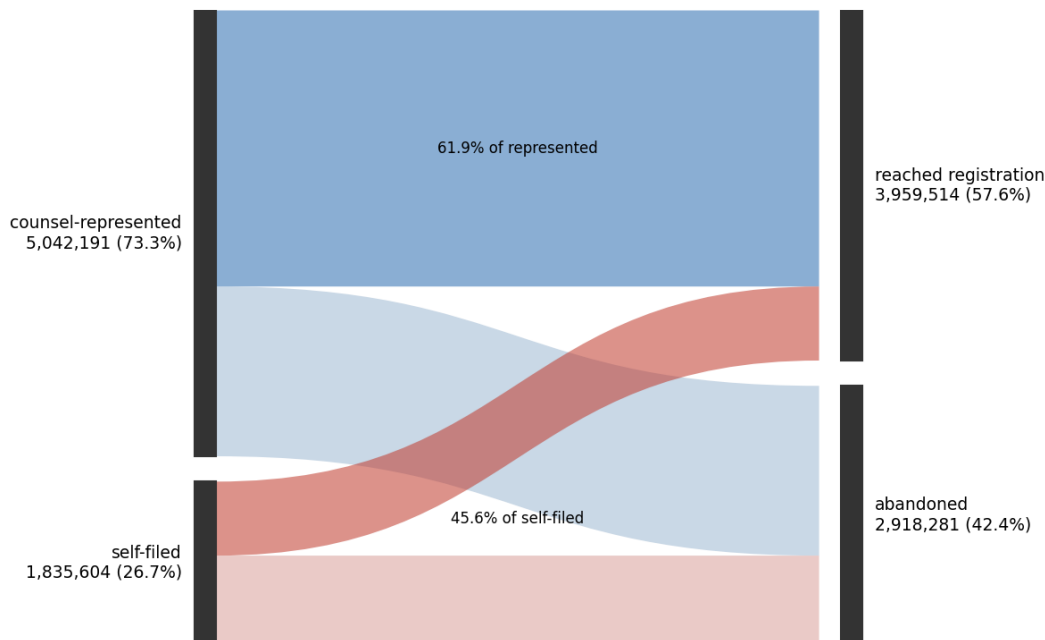


Figure S3: The registration step as a flow, unique serials filed 1995–2018 ($n = 6,877,795$), split by counsel of record on the filing. Represented applications reach registration at 61.9%, self-filed at 45.6%.

differently and fare differently:

	Share of filings	Reached registration	Median terms (009 / 035)
Represented	73.3%	61.9%	28 / 26
Self-filed	26.7%	45.6%	21 / 17

Self-filers write shorter descriptions (a median of 21 terms against 28 in the software class, 17 against 26 in advertising and retail), sit closer to their class’s usual language, and lean slightly further ahead of it. Because who drafts affects both the language and the outcome at once, the five-year-proof regressions control for counsel of record and report their estimates separately for the two groups. Figure S3 draws the full flow from filing to registration split by drafter.

S1.3 Office actions and refiling: applicants rewrite toward the class

An application that is abandoned is usually the end of that attempt: among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark again in the same class, and 1.5% by one that then registers. The refilings that do occur show what an applicant changes. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a different goods/services description the second time. To see which way the text moved, we score each of these registrations twice: once on the text of the abandoned first attempt, once on the text it registered with. Both self-filers and represented owners rewrite toward the class’s typical description: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16. Refilings that reuse the identical text move by

less than 0.01 at every position, so the convergence is in the writing. Who drafts changes the extent but not the direction: owners who hire counsel for the second attempt rewrite most often (82%) and their descriptions lengthen by 16%; counsel rewriting their own earlier draft rewrite least often (72%) and shorten by 4% (Figure S4). Lead barely moves once we account for the reference windows themselves shifting with the later filing date.

Figure S4 shows the goods/services description before and after a rewrite, for the 31,084 refilings in Section S1.3 whose text changed. A pair is an abandoned application and the same owner’s later registration of the identical mark in the same Nice class within six years; the earliest registered refiling is kept. Counsel of record is read from the prosecution history of each filing separately, so a pair can change representation between attempts. Atypicality is on the production scoring; length is the word count of the normalized description.

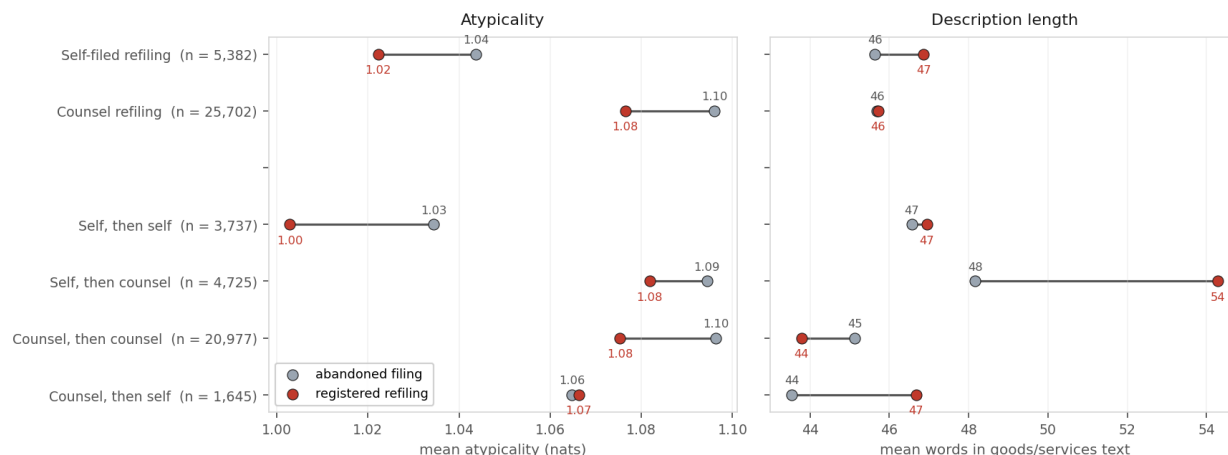


Figure S4: Rewritten refilings: mean atypicality and mean description length of the abandoned filing (grey) and of the registered refiling (red), by representation on the refiling and by the change in representation between the two attempts. Mean atypicality falls by 0.01 to 0.03 nats in every group except counsel-then-self, where it is unchanged. Length rises by six words where counsel is hired for the second attempt and falls by one where counsel rewrites counsel.

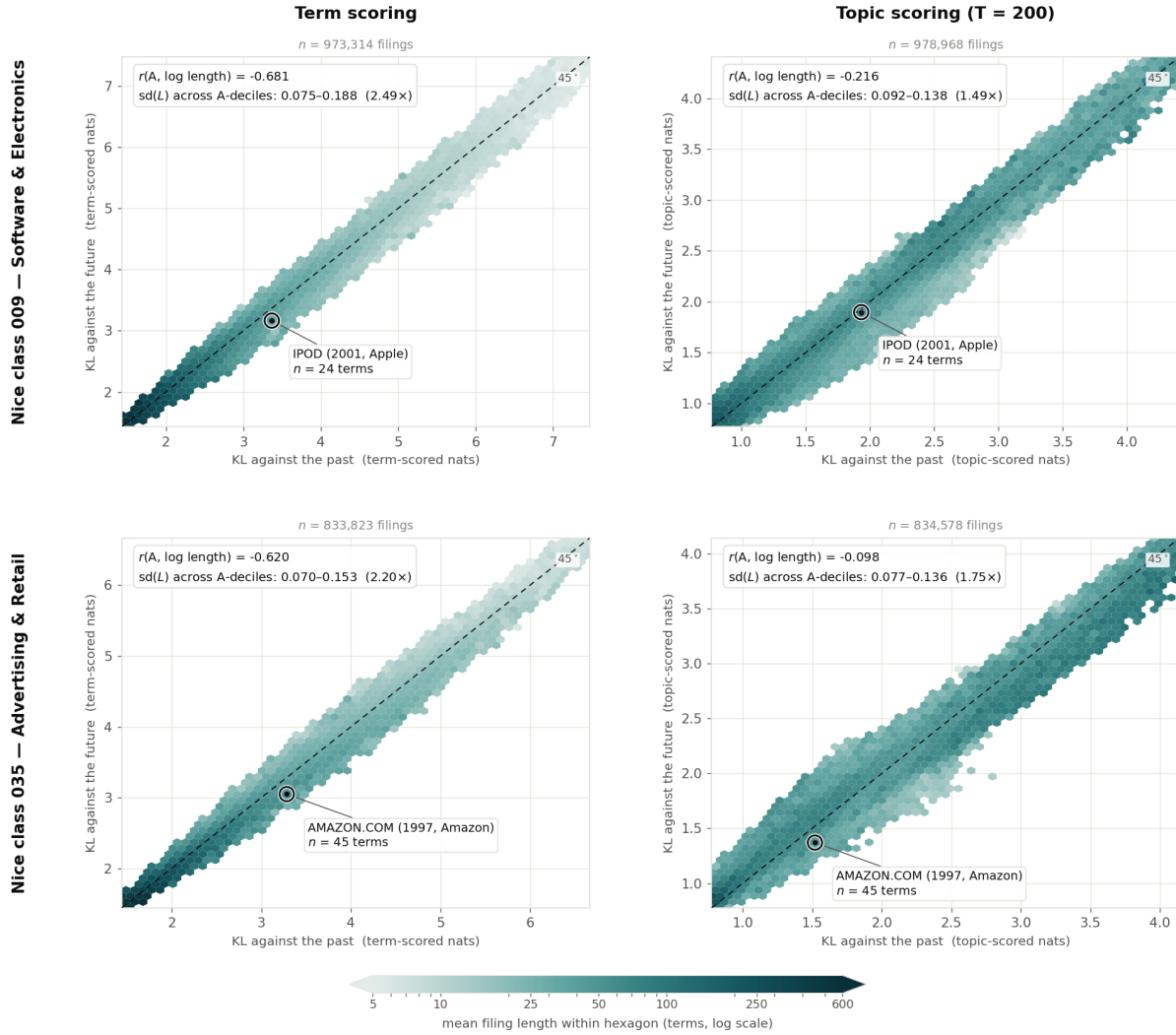
The means understate the convergence, because it runs in both directions. By fifth of the original’s atypicality, the change is +0.16, +0.14, +0.06, −0.06 and −0.39 nats from the most conventional fifth to the most atypical, and within a point of that profile in each of the four representation groups. Identical-text refilings change by +0.002 to +0.008 across the same fifths. On lead, the changed-text profile runs +0.046 to −0.072 and the identical-text profile +0.014 to −0.054; the difference is the rewriting, the rest is the reference windows moving with the later filing date.

S2 Representation: what terms resolve and themes do not

S2.1 Term scoring measures length; theme scoring does not

Figure S5 puts the two representations side by side over the same axes, with each hexagon coloured by the mean length of the filings in it. Under term scoring the colour runs from dark near the origin — a median of about 150 to 170 terms — to near-white at the far end, where the median is eight. That gradient is the measurement problem: a filing scores as atypical largely because it is short. Under theme scoring the same surface is a narrow band of roughly uniform tone. The theme

panels are drawn from the 200-theme build; at the production 50 themes the same check gives a correlation between theme-scored atypicality and log term count of between -0.10 and $+0.10$, against -0.64 under term scoring (article, Section 3).



Hexagons holding fewer than 25 filings are left blank; the retained hexagons still cover over 99.4% of filings in every panel.
 The colour scale is shared by all four panels. The axis ranges are not: each panel is boxed on its own 0.5th–99.5th percentiles, and term-scored and topic-scored KL do not share a scale — the comparison invited across columns is of shape and colour gradient, not of raw magnitude.

Figure S5: Mean filing length across the two axes, by representation and class, for filings 1995–2019 scored on per-filing reference windows (theme panels at $T = 200$). The horizontal axis is surprise against the preceding five years, the vertical axis surprise against the following five; a filing on the dashed diagonal is equally far from both, and distance below it is lead. Hexagons holding fewer than 25 filings are blank; the retained hexagons cover over 99.4% of filings in every panel. The colour scale is shared across panels; the axis ranges are not, because term-scored and theme-scored KL are not on a common scale, so only shape and colour gradient are comparable across columns. Both columns are drawn on the filings the per-filing windows score, which differ between representations by under one percent.

Each representation has the virtue the other lacks: theme scoring is independent of document

length, which term scoring is not, while words — the exhibit below counts word pairs as well as single words — show specific new wording, which theme scoring does not. We therefore use theme scoring for every estimate, because the length confound contaminates firm-level comparisons, and words for the worked examples, because the theme basis has no coordinate for the language the examples turn on.

S2.2 How three filings encode

Table S1 traces three filings through term scoring and theme scoring at $T \in \{50, 200, 500\}$ — three kinds of novelty, and how each representation handles them.

AMAZON.COM (1997, advertising/retail) is *combinatorial* novelty: every word is ordinary, but the bigrams assemble an invention (“computerized line,” “line search,” “search ordering”; class document frequencies 379, 134, 355). The filing’s text reads “on line” as two words, the standard spelling in 1997, before *online* lexicalized into a single token; the analyzer drops the stopword “on” and the bigram “line search” survives stopword removal. The novelty was necessarily combinatorial because the word for it did not yet exist. Term scoring sees these bigrams; under theme scoring at every resolution all of them are out of vocabulary and the filing reads as a media retailer that uses computers.

KOMBUCHA (1997, beer/soft drinks) is *lexical* novelty: the new thing arrives as new words (*kombucha* df 567, *saccharose* out of vocabulary even at the class level). The word *kombucha* is in the theme vocabulary but, with no kombucha-like theme, the filing loads on a generic beverages theme at every T (Table S1).

ETHEREUM (2018, software) is *wave* novelty: emergent vocabulary already diffusing (*blockchains*, *distributed computing*). It loads on generic software and services themes at every resolution, including $T = 200$, which does contain a cryptocurrency/blockchain theme; there, the eleven occurrences of *software* dominate the theme assignment.

Table S1: Top theme loading for three filings, by representation. Words show the novel content of all three; theme scoring shows none of it at any tested resolution. Loadings are computed with the model’s fitting tokenizer and can differ from the scoring-time loadings quoted in the article by about 0.01.

	AMAZON.COM (1997)	KOMBUCHA (1997)	ETHEREUM (2018)
novel content (term)	line search, search ordering	kombucha, saccha- rose	blockchains, dis- tributed computing
$T = 50$	video, computer, music (0.40)	beverages, fruit, drinks (0.64)	software, online, downloadable (0.33)
$T = 200$	entertainment, video, games (0.29)	beverages, drinks, coffee (0.60)	field, services, devel- opment (0.37)
$T = 500$	computer, software, data (0.22)	beverages, drinks, fruit (0.61)	software, online, computer (0.49)

The pattern across the row is the measurement point in miniature: theme scoring cannot represent combinatorial novelty (Amazon), dilutes lexical novelty (Kombucha), and majority-votes away wave novelty (Ethereum) even when a fitting theme exists. The mark-level outcome results agree under both scorings despite this blindness, while the firm-level results diverge; that combination is what identifies the firm-level results as artifacts of the lexical specificity that word scoring additionally picks up (article, Section 5.5).

S2.3 Arrangement channels: pair lead and recombination

Combinations can be scored in their own right, at two levels (construction detail in the public repository). The first is *pair lead*: whether the class later moved toward the filing’s arrangements of themes. In the software class it is penalised much as lead is (+2.3 points at the five-year proof, $t = 8.4$, while correlating with lead at only -0.18). But it does not generalize. Rebuilt corpus-wide (where the software class’s net contrast is +1.6), it is zero raw and +0.5pp net of lead and atypicality (owner-clustered $t = 4.2$), significantly positive in 13 of 43 classes and significantly negative in 3 — including -2.1 in advertising and retail — so we report it as a class-specific bound rather than a second construct. The second is word-level *recombination*: adjacent word pairs that are new to the class while both words are established there, novelty assembled from familiar components (Fleming, 2001; Uzzi et al., 2013). In the same class it is rewarded: the top fifth fails 2.8 points less with lead and atypicality held ($t = -10.4$), and it is uncorrelated with pair lead ($r = -0.01$). The construct sees neither arrangement channel, and where they act they run in opposite directions, so its blindness to combination is a bounded omission rather than a hidden bias in one direction.

S3 Burn-in

S3.1 Thin references near the start of the record

Burn-in is the span at each end of the corpus where the reference windows cannot be filled, so scores there reflect a thin reference rather than the text being scored.

A filing’s score depends on having enough same-class filings on each side of it. A full five-year window exists on both sides for filings made 1990 through 2020, and the production scoring runs 1995–2019, holding back five years at the start of that span, where references are thinnest, and one at the end; a filing is also scored only when each of its windows holds at least 500 filings. Figure S6 shows how reference density builds up at the start of the record. How far a thin reference distorts the score was profiled on an earlier build of the measure (word-scored, with calendar-year references): a thin past reference inflates past-facing surprise, and across the classes with enough volume to profile, the median distortion was about 2 nats for filings made in 1990 and fell quickly as the window filled. On that build’s own criterion, though, 18 of 44 classes stayed noisy past 1995, into 1996–2004, so estimates that lean on the earliest cohorts — the companion paper’s era comparisons, in particular — are read with that in mind.

S4 Robustness: what changes under other choices

Each result in Section 5 is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. This section reports what each of those choices costs, and in each case states the more complicated finding beside the simpler one the body uses.

S4.1 Representation and weighting

A scoring is fixed by two choices. The *representation* is what the language is turned into: a distribution over 50 fitted themes, or a distribution over the class vocabulary’s terms. The *weighting* is how days inside the reference window count: every day equal, or decaying by half every two years. Every reference is per-filing, anchored on the filing’s own date (Section 3). Four builds exist:

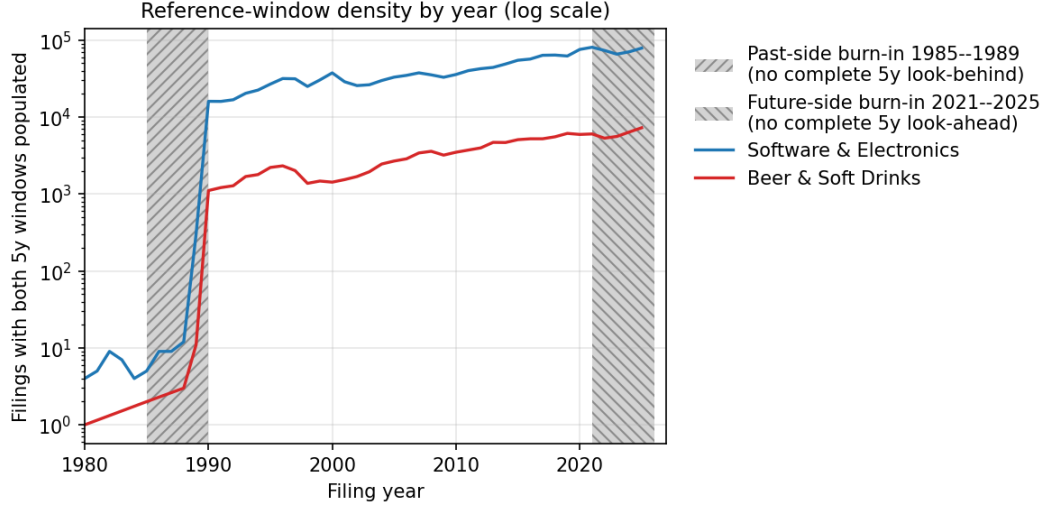


Figure S6: Reference-window density by year on a log scale, for two classes, from the earlier word-scored build. Hatched regions are the past-side and future-side zones where the five-year window is incomplete. The windows decide the score at the filing date; the proof outcome is read separately, at registration age six to seven.

Representation	Weighting	sd(lead)	sd(atypicality)
Themes	flat	0.102	0.698
Themes	half-life 2y	0.082	0.698
Terms	flat	0.155	1.101
Terms	half-life 2y	0.130	1.102

All four are compared on the 6.01 million filings every one of them scored. The first row is the production scoring.

Changing one choice at a time settles the ordering. Correlations of lead between builds differing in exactly one respect:

Choice varied	r (lead)	r (atypicality)
Weighting, flat vs half-life 2y (themes)	0.982	1.000
Weighting, flat vs half-life 2y (terms)	0.970	1.000
Resolution, 50 vs 200 themes ¹	0.636	0.799
Representation, themes vs terms	0.460	0.396

The weighting is close to irrelevant: whether recent language counts for more than distant language within the window barely moves the score, and it does not move atypicality at all to three decimals, which is what the sum-and-difference construction of Section 3 implies — atypicality is a property of the filing, and the weighting acts almost entirely on the small residual that is lead. Resolution and representation matter most: quadrupling the number of themes leaves the two builds agreeing on 40% of the variance in lead, and themes against terms on less than a quarter. A robustness check that varies only the weighting is therefore close to a restatement; one that varies the representation or the resolution is a real test, which is why the term-scored replication in §S4.10 and the sweep below are the ones the claims lean on.

¹The resolution row is computed on the five-year-proof sample rather than the 6.01 million-filing common sample of the other rows.

S4.2 Theme resolution

Fifty themes over a 62,168-term vocabulary is roughly one theme per industry; whether the findings depend on that coarseness is testable by refitting at higher T . Raising the theme count does not add coverage — the vocabulary is fixed, and a larger T re-partitions the same words — but it changes what counts as sitting far from an industry, and could change any sign. Re-fitting the model at four resolutions on the same stratified sample and the same analyzer, and re-scoring the software class under per-filing windows:

Themes	Proof contrast, lead	t	Proof contrast, atypicality	r with $T = 50$
50	+3.23pp	13.8	−14.01pp	—
100	+2.71pp	11.6	−13.69pp	0.690
200	+5.20pp	22.2	−13.45pp	0.640
500	+5.69pp	24.3	−14.49pp	0.641

Three readings, in descending order of how much weight they carry. The sign and the significance are untouched across a tenfold change in resolution, and so is atypicality, which stays within a point of −14pp throughout. The magnitude is not: it does not move monotonically in T , so no single resolution can be quoted as the software class’s effect, and the figure this paper reports at $T = 50$ sits at the low end of the range rather than the high one. (A fifth fit, at $T = 1000$, failed a training-convergence check and is excluded.) And agreement with $T = 50$ settles at 0.64 without decaying as T grows, which says the resolutions share a fixed core and differ in the remainder rather than drifting apart as the partition refines.

The corpus-wide contrast is steadier than the single class: scored for all 45 classes, it is +2.29pp at $T = 50$, +2.23pp at $T = 200$ and +2.61pp at $T = 500$. Pooling evidently averages away much of the per-class resolution sensitivity.

S4.3 Fitting noise at one resolution

The sweep leaves one question open. Scores at different resolutions agree at $r = 0.64$ to 0.69 on lead, and neighbouring resolutions agree no better than distant ones. That has two readings. The number of themes may genuinely change what is being measured. Or a theme-model fit may be unstable, with two runs settling in different local optima, in which case the sweep measures fitting noise and the resolution has little to do with it.

Holding T fixed and moving only the seed separates the two. A second fit at $T = 200$ on the same sample, the same vocabulary and the same number of passes, differing from the first in nothing but its random seed, agrees with it at $r = 0.79$ on lead and $r = 0.87$ on atypicality across 1.11 million scored software-class filings. The cross-resolution figures were 0.64 and 0.83. Most of what the sweep showed as resolution sensitivity is therefore present between two fits that differ only in the seed: for atypicality nearly all of it, for lead well over half. The resolutions are not measuring different things; each fit is a noisy draw on the same thing.

The estimate moves less than the scores do. On the identical sample, the five-year-proof lead contrast is +5.20pp under the first seed and +4.88pp under the second (SE 0.23 each), and the atypicality contrast −13.45 and −14.38pp. Two consequences follow. A single fit’s lead score carries measurement error of roughly a fifth of its variance, which attenuates every coefficient on lead toward zero; correcting for it would enlarge the estimates, not shrink them. And the seed explains most of the disagreement in per-filing scores between resolutions, but not the spread of the class-level estimates: the seed moves the software-class contrast by a third of a point, while

the resolutions in the table above span three. That spread reflects the scorings disagreeing about which filings are leading, and pooling across classes largely averages it out.

The same test at the production resolution gives a split answer. A second fit at $T = 50$, again differing only in its seed, recovers the themes themselves far more faithfully than at $T = 200$: matching each theme to its closest counterpart by the cosine of their word distributions, the median match is 0.74 at fifty themes against 0.54 at 200, and a quarter of the fifty themes match above 0.9. Per-filing scores are no more stable for it: the two fits agree at $r = 0.72$ on lead and 0.84 on atypicality across the same 1.11 million software-class filings, and the five-year-proof lead contrast is +3.23pp under the first seed and +3.89pp under the second (SE 0.23 each). A coarse model’s word lists are reproducible; which theme a borderline filing’s words are credited to is not, and that is where the per-filing noise comes from.

The substantive claim does not depend on any of this. The five-year-proof lead contrast on the 2002–2018 registrations each build scores runs +2.29, +1.95, +3.56 and +2.92pp across the four builds in the order tabulated above, and the atypicality contrast -4.39 , -4.43 , -3.63 and -3.73 pp. Every sign holds; term scoring returns the larger lead contrast, so the headline estimate is the conservative representation.

S4.4 Decayed reference windows

Inside the reference windows every day counts the same. The alternative weights recent language up, with a half-life of two years inside the same 1,826-day truncation, so that a term the class adopted last month is more familiar than one it adopted four years ago. Rescoring the corpus that way changes the scores little and the estimate somewhat. Flat and decayed lead correlate at 0.980 on the 3.10 million five-year-proof registrations and atypicality at 1.000 to three decimals. On the identical registrations, with quintiles cut within class and registration year, the top-minus-bottom proof contrast on lead is +2.29pp under the flat window and +1.95pp under the decayed one (SE 0.09 each); on atypicality it is -4.39 and -4.43 pp. The decayed window therefore returns a lead penalty about a seventh smaller, with the same sign, the same precision, and the same profile. The flat window is kept in the body because it is the simpler statement; a reader who prefers recent language to count for more should take +1.95pp as the estimate.

S4.5 Window width

The window is a parameter of the measure, not a property of it, and the five years used throughout was fixed early in the project rather than tuned to the present construction. To test it, we re-score the software class at several widths under per-filing references and judge each width by the five-year-proof quintile contrast it produces: +2.02pp at two years, +2.20 at three, +3.23 at five and +3.37 at seven, each on an identical sample of 452,911 registrations and each with a standard error of 0.23pp. Signal rises steeply from two to five years and is flat between five and seven: the +1.03pp gain from three to five is well outside the sampling error, the +0.14pp from five to seven is not. A ten-year window returns +2.89pp, but that figure is not comparable to the others — requiring both windows to lie inside the corpus costs 22% of the scored filings and 13% of the proof sample, dropping the earliest and latest registration years, which are also the years whose contrasts differ most. The apparent decline at ten cannot be separated from that change in composition and is not evidence about window width. On the range where the sample is fixed, seven years is nominally the peak and five is within noise of it while retaining more of the corpus, so five is defensible rather than optimal. The sweep is on one class and one outcome.

S4.6 Mixing the two windows

“Leading” is relative to a reference, and the two references need not be the same width. A short past and a long future make lead a measure of foresight — whether the filing resembles where the class was heading over the next seven years more than what it wrote in the last three. A long past and a short future make it a measure of rupture — whether the filing has broken with what the class had established over seven years, judged against only the next three. We score every class on the production representation and per-filing windows at $W \in \{3, 5, 7\}$ years on each side, and evaluate all five variants on identical samples. In the table, the proof columns are computed on the 3.10 million registrations 2002–2018, with quintiles cut within class $\times$ registration-year; the registration columns on the scored filings 1995–2018, with quintiles within class $\times$ filing-year:

Variant	Past, future (years)	Five-year-proof failure		Registration	
		Q5–Q1 (pp)	classes > 0	Q5–Q1 (pp)	Q3 – tails
Foresight	3, 7	+1.65 (0.09)	31 of 44	+1.09	+0.50
Symmetric	3, 3	+1.69 (0.09)	—	+0.79	+0.13
Symmetric	5, 5	+2.29 (0.09)	33 of 44	+0.71	+0.28
Symmetric	7, 7	+2.58 (0.09)	—	+0.48	+0.65
Past-rupture	7, 3	+3.00 (0.09)	37 of 44	+0.01	+0.33

The symmetric five-year row is the paper’s FE-only estimate in Table 3, reproduced here to the decimal, which fixes the scale. The penalty rises monotonically along the mix, from +1.65pp under foresight to +3.00pp under rupture, and rupture is the more pervasive variant, positive in 37 of 44 classes against 31. In software the two are +1.82 and +5.07pp. So the five-year proof punishes distance from the class’s established language more than it punishes resemblance to its coming language; but foresight carries a penalty of its own, more than half the symmetric one, and the separation is a matter of degree rather than kind.

Registration is a different matter. On the signed axis the top-to-bottom contrast is at most a point on a 57% base and the interior-minus-tails depth at most two thirds of a point, under every mix. The examiner’s response to lead, on this scoring, is close to nil.

S4.7 Without conditioning on grant

Figure S7 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by ~ 2017 and face at least one Section 8 proof by the record’s end in April 2026; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two proofs). Panel A is the unconditional composite, the probability that a filing both registers and passes the proof, measured over all filings including the never-registered. It is nearly flat in lead because two opposed gradients cancel: registration rises in lead (61.3% to 63.2%) while proof survival falls (44.2% to 40.8%); the caption carries the quintile figures. Conditioning on grant is what separates them. Panel B shows the conditional proof outcome on the same filings, and it is U-shaped rather than inverse-U (depth -2.11 pp), with a clearly penalised leading tail consistent with the cleaner single-proof cohort of Figure S8.

S4.8 Toggled filters, windows and parameters

Each analysis in this paper and in the companion paper was rerun under its natural parameter toggles, on the identical data. Nothing reverses a headline claim; three auxiliary readings shift.

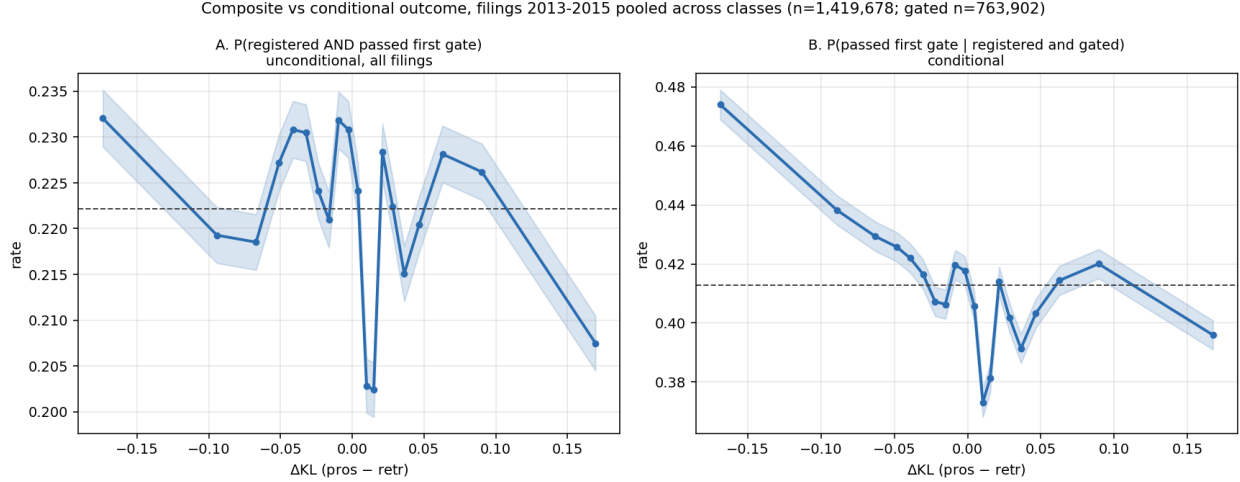


Figure S7: Filings 2013–2015 pooled across classes, theme-scored on per-filing reference windows ($n = 1,419,678$; proof-facing subset $n = 763,902$). *A*: the unconditional composite $P(\text{registered and passed the five-year proof})$ over all filings. *B*: $P(\text{passed} \mid \text{registered and facing the proof})$ on the same filings. The unconditional view is nearly flat in lead (22.4% to 22.1% across quintiles); conditioning on grant is what exposes the gradient.

Variant	Baseline $\rightarrow$ variant	Reading
Gate window 3.5–9.0 / 4.5–8.0	lead +2.29 $\rightarrow$ +2.29 / +2.28pp	window immaterial
Registration cohorts 2002–09	+2.29 $\rightarrow$ +0.98pp	the era swing
Registration cohorts 2010–18	+2.29 $\rightarrow$ +3.11pp	penalty concentrated late
Atypicality, both cohort halves	–4.39 $\rightarrow$ –4.55 / –4.28pp	stable
Internet pattern, narrow core	goods web gap 0.0 $\rightarrow$ +2.0pp	divergence holds
Convergence arrival 1% / 4%	early $\rightarrow$ late +4.4 $\rightarrow$ +1.6	+5.9 $\rightarrow$ +1.8 / +3.8 $\rightarrow$ +2.3
Surge doubling 1.5 $\times$ / 3 $\times$	in-surge net +1.3pp	+1.2 / +0.9pp
Wave entry window 4y / 8y	order flat	flat under both
Recombination floor 2 / 10	net –2.8pp	–1.2 / –3.6pp
Pair-lead presence 0.05	net +2.3pp	+3.1pp
Combination measures, class 035	009 values	recomb –3.8; pair-lead –1.4
Ladder, counsel / self debuts	reporting lead –0.19pp	–0.26 / –0.03pp
Ladder, 2009–13 / 2014–18	reporting atyp. +0.11pp	0.00 / +0.19pp
Excluding duplicate-text filings	lead +2.29; atyp. –4.39pp	+3.05; –5.68pp
Excluding Manual-scale texts only	same	+2.64; –5.05pp

Three of these change what a reader should carry. The lead penalty at the proof belongs to the 2010–2018 cohorts; the 2002–2009 half carries a third of it, which is the era swing the companion paper documents, seen from the other side. The ladder’s gradients belong to counsel-represented debuts — among self-filers the reporting gradient is a null — and its atypicality gradient to the 2014–2018 half. And the pair-lead penalty, +2.3pp in the software class under every parameter tried there, is –2.1pp in advertising and retail on the corpus-wide construction, and pooled across 43 classes is zero raw and +0.5pp net of lead and atypicality, so it is reported as a bound for one class rather than a general fact.

S4.9 Lead is a small residual

Past-facing and future-facing KL are highly correlated however scored. On the complete class-009 corpus under the production scoring they correlate at $r = +0.988$ ($n = 1,109,643$). The lead L is therefore a small residual of two large, stable quantities: both levels have a standard deviation near 0.70 nats about a mean of 1.46, while L has a standard deviation of 0.109 — about a sixth of either level’s spread. Only 1.2% of the two levels’ combined variance survives into their difference. That is what makes L sensitive to anything that inflates the levels; under term scoring the dominant such thing is document length, and Section 3 quantifies the coupling under both representations. The consequence for lead is direct: the standard deviation of L rises 2.2–2.5-fold across deciles of A under term scoring and 1.5–1.8-fold under theme scoring, so a unit of lead means roughly the same thing everywhere only in the theme representation. §S2.1 shows this as a length surface over the two axes.

One qualification: “flat” means no monotone length gradient rather than constant: class 035 under theme scoring is mildly U-shaped in A . The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.42 on the 6.01 million filings both score). That is what gives their agreement on the mark-level patterns evidential force, and their disagreement on the firm-level patterns diagnostic force (Section 5.5). Across $W \in \{3, 5, 7\}$, 6.9–10.4% of filings flip sign. Per-filing noise attenuates at every aggregation used: firm-year means, token pools, theme prevalence trajectories.

Alignment with expert innovation judgment is at most suggestive. Firms on BCG/MIT “most innovative” lists sit +0.09 standard deviations above the panel mean under term scoring ($p = 0.22$), +0.14 under theme scoring ($p = 0.07$), and +0.08 at $T = 200$ ($p = 0.20$). The matched sample is 41 firms, which is too small to carry weight in either direction, and the paper draws no inference from it. Patent counts remain uncorrelated with lead on the same panel (Pearson +0.010).

S4.10 Term scoring as a cross-check

The snapshot cross-check on the single one-proof-elapsed cohort (registrations 2016–2018, terminal status 710 versus 701–705/800) agrees with the event-dated estimates and is scoring-robust: -5.4 pp survival across quintiles under theme scoring and -5.7 pp under term scoring on the identical sample (Figure S8). Under term scoring, proof survival *rises* in the raw KL level ($+6.5$ pp past-facing, $+7.6$ pp future-facing): atypical text survives the five-year proof better while leading text survives it worse — the same two-axis separation as Section 5.4, at the mark level.

S4.11 Functional form

Four shapes were fitted to each profile by weighted least squares on 40 equal-count bins (Figure S9): linear, quadratic, an exponential $a + be^{cx}$, and a free-vertex symmetric form $a + be^{c|x-x_0|}$ meant to catch a U directly. Selection is by AIC.

Outcome	Axis	R^2 line	R^2 curve	Δ AIC	Shape selected
Registration	atypicality	0.38	0.92	77.1	curve, peak at +1.04
Registration	lead	0.08	0.17	1.9	(quadratic; weak)
Use-proof	atypicality	0.74	0.74	0.0	linear
Use-proof	lead	0.44	0.55	4.3	V, vertex -1.50
SEC reporting	atypicality	0.74	0.84	15.2	curve, peak at +2.17
SEC reporting	lead	0.13	0.59	26.3	V, vertex +0.30
listed only	lead	0.19	0.46	12.2	V, vertex +0.41
reporting, not listed	lead	0.07	0.67	37.2	V, vertex +0.24

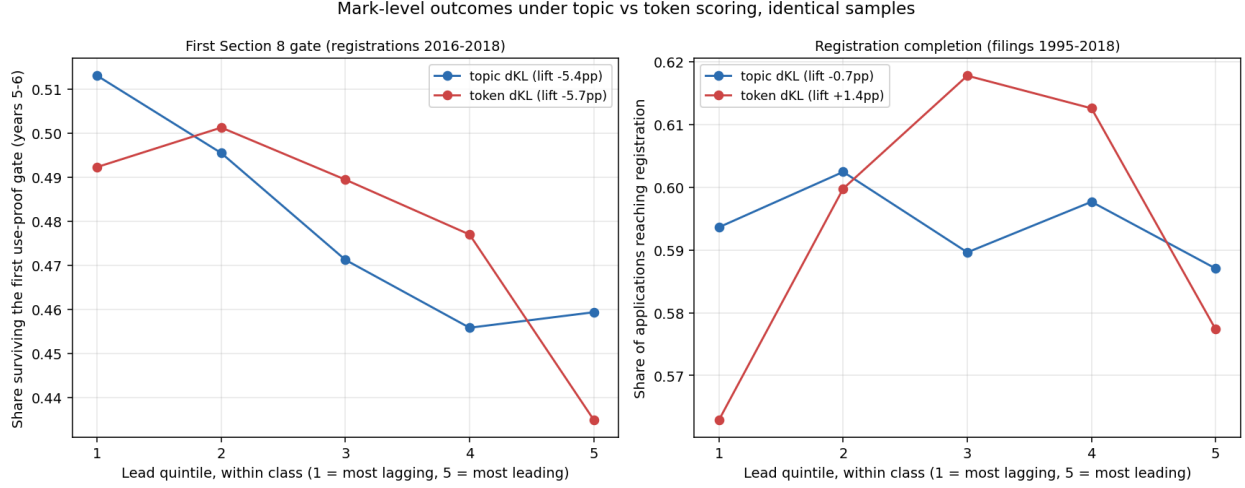


Figure S8: The two mark-level outcomes under theme and term scoring on identical samples, pooled across classes, by within-class lead quintile (1 most lagging, 5 most leading). *Left*: the share of registrations 2016–2018 that survived the five-year proof of continued use at years five to six, read from terminal status codes, where the two representations agree closely. *Right*: registration completion (filings 1995–2018), where they do not: the inverse-U in the signed axis is present under term scoring (depth +0.048) and absent under theme scoring (depth –0.001). The proof result is scoring-robust; the registration curvature on this axis is not, and the registration analysis reports only the unsigned version.

The winning form is a free-vertex $a + be^{c|x-x_0|}$. On the atypicality rows it is a genuine curve, with fitted exponents between 0.5 and 1.2 — a hump that peaks about one standard deviation above mean atypicality at registration and about two above it at SEC reporting. On every lead row the exponent collapses to the order of 10^{-4} , which turns $e^{c|x-x_0|}$ into $1 + c|x - x_0|$: what wins there is a V , linear in distance from a vertex. A V beating a parabola says the turn is sharper than a quadratic allows, not that anything grows exponentially.

Vertices are in standard deviations of the axis across filings.

The two rows for SEC reporting are the outcome Section 5.4 and Table 4 actually run on; the listed and reporting-only splits below them show the same shape on each half, which is why the split does not change the reading.

AIC here is computed on 40 binned means, not on the millions of underlying filings, so it ranks shapes rather than testing them. The 77 at registration and the 12 to 36 at the public-reporting steps are decisive; the 1.9 at registration on lead and the 4.3 at the five-year proof are weak, and those two rows are better read as “no worse than linear” than as curvature.

The non-monotone profiles on signed lead decompose into parts of different symmetry (Figure S10). Each ventile’s deviation of the listing rate from the base is split, by sequential counterfactual prediction, into the part atypicality composition alone predicts, the part exact-duplicate text adds, a linear lead tilt, and a residual. The symmetric U is carried about half by atypicality composition — the spread of L rises with A , so both tails of the lead axis hold atypical filings and the middle holds typical ones — and about half by a residual that is itself symmetric in $|L|$: the same conditional lead magnitude effect Table 4 estimates at +0.067pp per standard deviation. Duplicate-text composition adds almost nothing once atypicality is held, and the linear tilt is an order of magnitude smaller than either symmetric part (–0.07pp per nat). The fitted vertices above follow from this arithmetic: a symmetric profile plus a negative tilt has its minimum shifted right

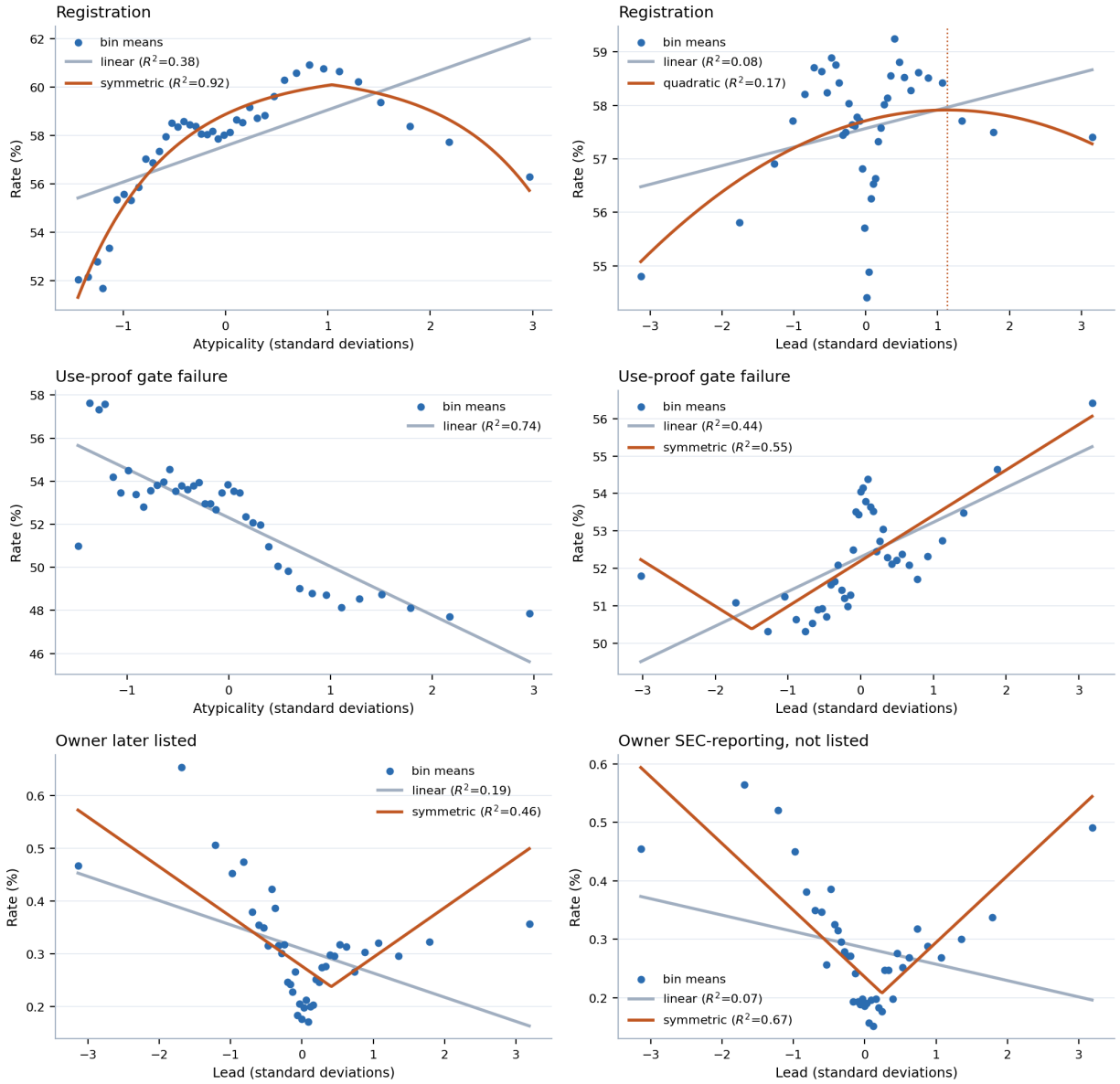


Figure S9: Binned profiles at the three steps, with the linear reading the paper's contrasts assume and the shape selected by AIC where that is not linear. Forty equal-count bins; the dotted line marks the fitted vertex. The bottom row is the case that matters: a linear fit through the public-reporting profile slopes downward while the data turn up at both tails.

of zero, and a survival hump its maximum shifted left, which is where the fits put them (+0.24 to +0.41 standard deviations for the reporting minima; the survival vertex, at -1.5 , comes from a weak fit, $\Delta\text{AIC } 4.3$).

Excluding text-duplicate filings strengthens rather than weakens every headline gradient. Of the five-year-proof sample, 47% of registrations share their exact normalized description with at least one other filing; dropping them raises the lead contrast from +2.29 to +3.05pp and the atypicality contrast from -4.39 to -5.68 pp, and dropping only the Manual-scale texts (clusters of 100 or more identical filings) gives +2.64 and -5.05 pp. Identical texts carry identical scores and cluster at the typical end of the axis, so they dilute both gradients; neither is created by them.

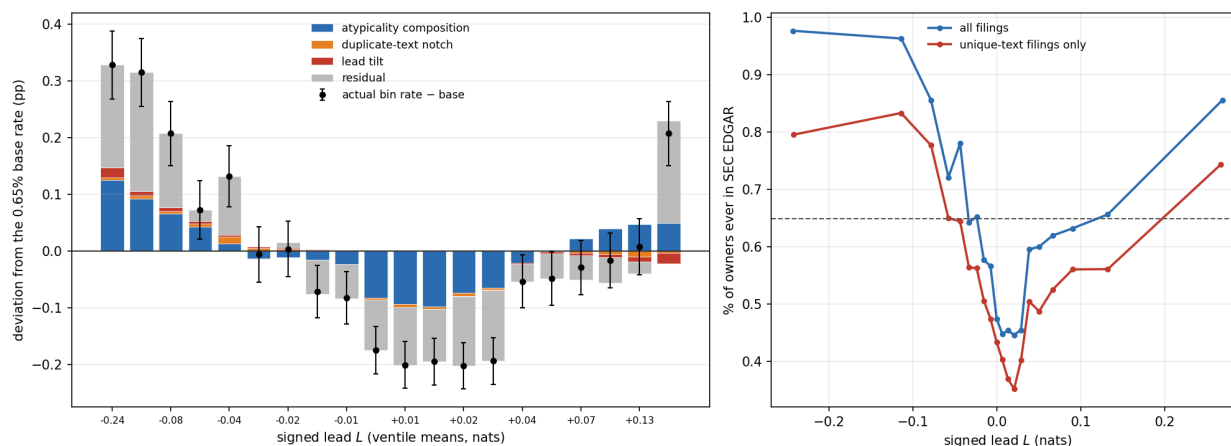


Figure S10: The signed-lead listing profile taken apart. *Left*: each ventile’s deviation of $P(\text{owner ever in SEC EDGAR})$ from the 0.65% base, on 2.07 million registered debuts 1995–2018, split into the part predicted by atypicality composition alone (50 quantile bins of A), the part added by an exact-duplicate-text indicator, a linear tilt in L fitted to the remainder, and the residual; bars stack from zero, dots are the actual deviations with 95% intervals. *Right*: the same profile on all filings and on filings whose normalized description text is unique in their class; the central dip survives, and deepens, without duplicated text.

S4.12 Listing against reporting

“Appears in SEC records” mixes two populations: companies whose shares trade, and entities that file with the Commission without ever listing — private issuers, funds, subsidiaries registering securities. Splitting the 19,889 matched owner strings on whether their CIK carries a ticker in the Commission’s own company-ticker file gives 10,556 listed against 9,333 that file financial statements without one.

It does not change the finding. Among registered debut owners whose debut filing is scored, 0.310% later appear as listed companies and 0.285% as reporting non-listed ones, and both profiles are U-shaped in lead with vertices at +0.41 and +0.24 standard deviations. The listed profile is slightly flatter and noisier, which is what the smaller and more selected outcome predicts. Whatever the signed axis is picking up at this margin, it is not an artifact of counting non-operating registrants as commercial successes, and it is not specific to the act of listing either.²

²Two limits on the split. The ticker file is a current snapshot, so a company that listed in 2003 and was acquired in 2011 appears in the non-listed tier; the tiers are therefore “currently listed” rather than “ever listed,” which will misclassify older successes downward. And the base rates here are computed on the appendix’s own panel, which

S4.13 Filings are not independent draws

Two dependences run through the corpus. Filings can share an exact position, because identical text carries identical scores. And outcomes cluster within owner: the within-owner correlation is 0.38 for proof survival pooled, and it is mechanically 1.0 for public listing, since listing is a property of the owner and a firm with hundreds of marks contributes hundreds of identical successes. The regression tables are unaffected, because the proof regressions cluster errors on normalized owner and the firm-level specifications carry one observation per owner. The raw quintile contrasts of the companion paper — the era, theme, internet and surge tables — are not: their two-sample standard errors ignore the owner clustering, and their t -statistics overstate precision by roughly a factor of two, which the text of those sections carries where the margins are close. Binning filings without the correction likewise overstates how much structure the corpus contains. On a positional grid, correcting cell errors for the clustering design effect — the factor by which within-owner correlation inflates the variance of a cell mean — cuts apparent excess dispersion by roughly a quarter on the proof outcome and by half on listing. Displays of the vocabulary plane are descriptive here for that reason; and since the near-zero pooled correlation between A and L reported in Section 3 does not survive cell by cell within class, per-cell contrasts should not be read as within-class effects.

References

- Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1), 117–132.
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472.

conditions on the debut filing carrying a score and so is smaller than the body's sample.